physica status solidi pss a
www.pss-a.com

The Impact of Contact Position on the Retention Performance in Thin-Film Ferroelectric Transistors

*Qiusong Chen, Dong Lin, Qinhao Wang, Jiang Yang, Juan E. Sanchez, and Guodong Zhu**

Here, the simulation of the physical fields in an n-type ferroelectric thin-film transistor (FeTFT) with a bottom contact bottom-gate configuration is demonstrated. In the off-state, the intensity of the lateral electric field in the space charge region near the edge of the drain electrode is several times higher than the coercive field of the ferroelectric film, pinning the direction of polarization near the drain electrode along the channel. While the FeTFT switches from off-state to on-state, the vertical component of polarization near the drain edge is stronger than other parts near the channel. This enhances the depolarization field in this area so that the retention performance of the on-state is worse than the off-state. These results and methods benefit FeTFTs and are also valuable for floating-gate memory and light-emitting field-effect transistors.

1. Introduction

Ferroelectric memory, typically realized using the ferroelectric thin-film transistor (FeTFT), has progressed toward high endurance,[1,2] multilevel storage,[3–5] and flexible device fabrication.[6,7] The application of 2D materials, such as black phosphorus, MoS_2, and graphene, provides more choice for tunable direct bandgap and high carrier mobility semiconductors.[8–11] All of these advancements make FeTFT became a possible candidate for the next-generation electronic devices, such as negative capacitance field-effect transistors[12,13] and ferroelectric random-access memory.[14] But, a hurdle for its practical implementation is its relatively poor retention performance.

The depolarization field is the electric field in the ferroelectric layer when the device is in standby condition.[15] For all possible reasons, this field is regarded as the main factor in the retention performance of FeTFTs.[15,16] Huang et al.[17] suggested that the retention performance of p-type FeTFT in the off-state was worse than that in the on-state, due to the larger depolarization field. Wurfel et al. reported that the depletion layer at the inversion regime increased the depolarization field.[16] Pan and Ma reported that both the depolarization field and gate leakage were responsible for memory window loss in different time regimes.[18]

While these investigations revealed the basic retention mechanisms of FeTFTs, they assumed uniform physical properties in the ferroelectric layer. However, the physical details near the source and drain electrodes are also important. The electric field at the edge of the electrode can be up to 8 MV cm^{-1} in the off-state.[19] The intrinsic coercive field predicted by Landau–Ginzburg's theory can reach 2–10 MV cm^{-1}.[20,21] Localized nucleation of domains with reversed polarization makes the real value of coercive field typically many orders of magnitude lower than the theoretical prediction, 0.5 MV cm^{-1} for fluoride polymer and 1 MV cm^{-1} for HfO_2.[20,21] The fields in the source and drain junctions are several times higher than the coercive field in a real ferroelectric film.

Based on the Maxwell–Faraday equation, the tangential component of the electrostatic field is continuous along the interface. If the ferroelectric layer adjoins the source and drain electrodes, as in bottom contact bottom gate (BCBG) and top contact top gate architectures, this junction field will affect the polling details of FeTFT, and thus determine the depolarization field and retention performance.

Experimental approaches have investigated the distribution of polarization in the ferroelectric layer and displacement transients in the gate current[22–27] These methods expose the overall states of polarization distribution or get information at the surface. However, they do not make a detailed analysis of the electric field around the electrodes.

To investigate the effects of physical fields near the edge of the electrode, the first step would be to develop a method to explore the interaction between carrier conditions in the semiconductor layer and polarization distribution in the ferroelectric layer.

In this work, we modeled the distribution of physical fields in an n-type FeTFT with a BCBG architecture, using technology

Q. Chen, Q. Wang, J. Yang, G. Zhu
The Department of Materials Science
Fudan University
220 Handan Road, Shanghai 200433, China
E-mail: gdzhu@fudan.edu.cn

Q. Chen
School of Physics and Electronic Science
Guizhou Education University
115 Gaoxin Road, Guiyang 550018, China

D. Lin
The College of Information Engineering
Jimei University
Xiamen 361021, China

J. E. Sanchez
DEVSIM LLC
PO Box 50096, Austin, TX 78763, USA

The ORCID identification number(s) for the author(s) of this article can be found under https://doi.org/10.1002/pssa.202100408.

DOI: 10.1002/pssa.202100408

physica status solidi a
www.pss-a.com

computer-aided design (TCAD) device simulation software. TCAD is a finite volume method, which has been used extensively in the integrated circuit industry to analyze circuit behavior at a microscopic level.[28] The electrical details near the edge of drain contact and its impact were analyzed quantitatively. We explain the correlation between retention characteristics with electrode details. To the best of our knowledge, the effect of this junction field has not been previously presented in the literature.

2. Device Architectures and Numerical Details

2.1. Device Parameters

The configuration of the device simulated in this work was an n-channel thin-film FeTFT with a BCBG configuration and is shown in **Figure 1**a. For such devices, the conventional thicknesses of ferroelectric layers are 200–500 nm,[26,29] and semiconductor 20–80 nm,[29–31] respectively. In this work, we set 200 nm for the ferroelectric layer and 40 nm for the semiconductor layer. Being a 2D simulation, we assume a channel width of 1 mm. Additional geometric details are listed in **Table 1**. It should be noted that this thickness is not suitable for up-to-date thinner semiconductors.[32–34] For those 2D materials, their energy bands are different and more sophisticated than the conventional thicker or even bulk materials. We will analyze this issue in our future work.

In this work, we assume the source and drain electrodes are Ohmic and the channel is set to the Fermi level of the gate electrode (E_G) equal to the intrinsic Fermi level (E_{F_i}) of the undoped semiconductor (Figure 1b). The semiconductor layer in the FeTFT is simulated without any consideration of charge trap mechanisms and degradation effects. The parameters for our simulation are based on silicon as the semiconductor layer, and P(vinylidene fluoride-trifluoroethylene 70:30) (P(VDF-TrFE)) as the ferroelectric layer.[35,36] The device simulation parameters are listed in Table 1.

Table 1. The device parameters used in the simulation.

Symbol	Quantity	Value
L_G	Length of gate contact	1000 nm
L_D, L_S	Length of source, drain contacts	100 nm
T_S	Thickness of semiconductor layer	40 nm
T_F	Thickness of ferroelectric layer	200 nm
T	Device temperature	300 K
ϵ_{rf}	Relative permittivity of ferroelectric material	12
ϵ_{rs}	Relative permittivity of semiconductor material	4
P_s	Saturation polarization	12 μC cm^{-2}
P_r	Remnant polarization	10 μC cm^{-2}
μ_e	Electron mobility	10^7 μm^2 (Vs)$^{-1}$
μ_p	Hole mobility	10^5 μm^2 (Vs)$^{-1}$
V_S	Electron saturation velocity	10^{11} μm s^{-1}

2.2. Numerical Method

For simulation, we utilized DEVSIM, an open-source TCAD device simulation software.[37] It employs symbolic model

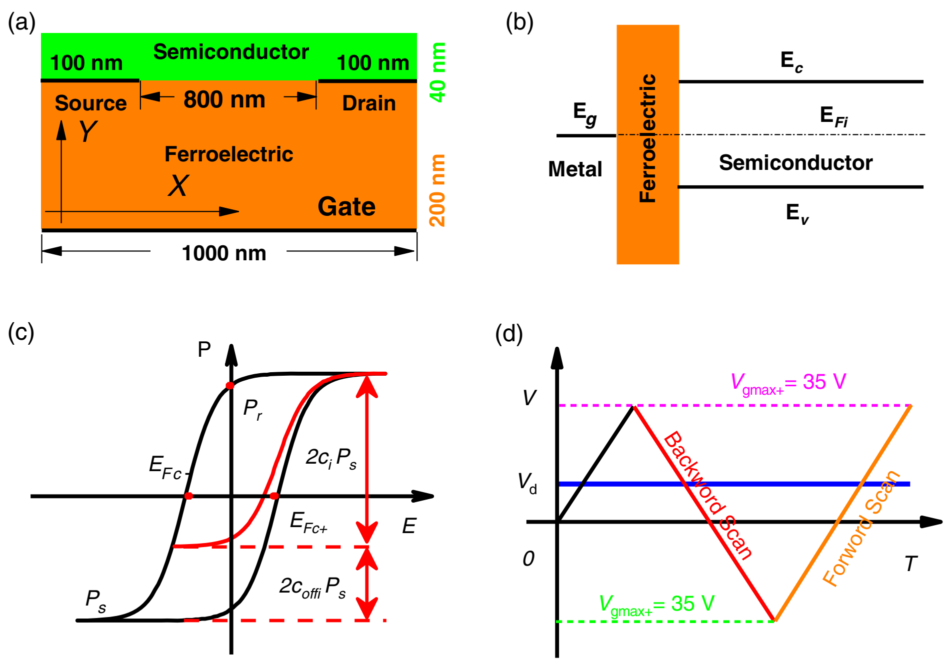

Figure 1. a) The schematic representation of BCBG FeTFT. b) The energy diagram of gate metal, ferroelectric insulator, and silicon semiconductor. E_G is the Fermi level of gate metal. E_c, E_V, and E_{F_i} are conduct band, valence band, and intrinsic Fermi level of semiconductor material, respectively. c) The curves of hyperbolic tangent function to simulate completely (black) and partly (red) polarized ferroelectric hysteresis loops. d) The diagram for the gate and drain voltage in the simulation process.

evaluation to simulate physical phenomena using continuum methods. The simulation mesh structure was generated by Gmsh.[38] The Poisson, continuity, and drift-diffusion transport equations were solved on a 2D mesh. The polarization of the ferroelectric material was simulated by a widely accepted empirical formula of the hyperbolic tangent function.[36,39–41] The polarization of the ferroelectric material as a function of the electric field is explicitly considered.

In electromagnetic theory, Gauss's law includes the polarization, $\overleftarrow{P}$, of a ferroelectric material and reads

$$\nabla \cdot (\varepsilon_r \varepsilon_0 \overleftarrow{E} + \overleftarrow{P}) = \rho \tag{1}$$

where ε_0, ε_r, and $\overleftarrow{E}$ are vacuum permittivity, relative permittivity, and electric field vector, respectively.

The charge density ρ is

$$\rho = q(p - n + N_D) \tag{2}$$

where q is the elementary charge and p, n, and N_D are hole density, electron density, and doping density.

The polarization $\overleftarrow{P}$ of the ferroelectric material non-linearly depends on the electric field, $\overleftarrow{E}$. The polarization vector $\overleftarrow{P}$ is split into its components along the three main axes of the mesh coordinate system, and each component reads[36,39–41]

$$P_i = P_s c_i \tan h[\omega(E_i - E_{F_c})] + P_s c_{\mathrm{off}_i} \tag{3}$$

where $i \in (x, y, z)$, $\overleftarrow{P_i}$ is the polarization in the i-direction

$$\omega = \ln\frac{(P_s + P_r)}{(P_s - P_r)} / (2E_{F_C}) \tag{4}$$

E_i is the electric field in the i-direction, P_s, P_r, and E_{F_C} are saturation polarization, remnant polarization, and coercive field, respectively. $c_{\mathrm{off}_i} \in (-1, 1)$ and $c_i \in (0, 1]$ are two parameters to describe the history dependency property of polarization hysteresis in the i-direction. From Equation (3), c_{off_i} is consistent with the polarization while E_i at E_{F_C}. c_i, c_{off_i}, and E_{F_C} are marked in hysteresis loops of Figure 1c.

The relationship between c_i and c_{off_i} is

$$c_i \cdot \mathrm{sign}(E_{F_C}) + c_{\mathrm{off}_i} = \mathrm{sign}(E_{F_C}) \tag{5}$$

where the sign function returns 1 while the electric field is increasing in i-direction, and returns −1 while the electric field is decreasing in the same direction. For the completely polarized ferroelectric hysteresis loop, the coefficient c_i is 1 and c_{off_i} is 0 (black in **Figure 2**). But for the partly polarized hysteresis loop, c_i is lesser than 1 and c_{off_i} is a nonzero value (red curve in Figure 1c). The actual value of c_i and c_{off_i} is calculated from Equation (3) to make the hysteresis loops coherent at the turning point of the electric field. Their values were iterated only when the sweep direction of the electric field changes so that we can track the changes of the polarization field through their values.

2.3. Operation Details for Transfer Characteristic Simulation

To reveal the effect of the electrode on FeTFT performance, the transfer characteristic measurement process was simulated. The initial equilibrium state was calculated by setting the polarization to 0. The transfer curves were simulated under a quasi-steady state.

For the whole simulation process, we kept drain bias at 5 V and source contact grounded. Then the gate bias (V_g) was swept from 0 V to a positive value ($V_{\mathrm{gmax}+}$). This will let the FeTFT go to the strong inversion regime. Second, the backward scan swept V_g from $V_{\mathrm{gmax}+}$ to a negative value ($V_{\mathrm{gmax}-}$). The magnitude of $V_{\mathrm{gmax}+}$ and $V_{\mathrm{gmax}-}$ are three times larger than the corresponding coercive voltage of the ferroelectric layer to confirm a full polarization reversal. As the coercive voltage is 10 V for 200 nm ferroelectric layer, we set $V_{\mathrm{gmax}+}$ to 35 V and $V_{\mathrm{gmax}-}$ to −30 V. The third step was forward scan which swept V_g back to $V_{\mathrm{gmax}+}$. The operation details of V_g and V_d are illustrated in Figure 1d.

The whole transfer loop of drain current verses V_g for the FeTFT is exhibited in Figure 2a. The threshold voltages for

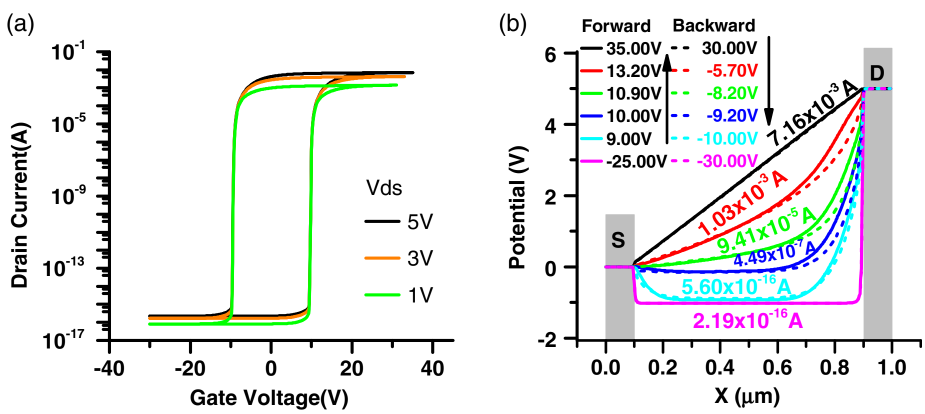

Figure 2. a) Transfer curves of simulated FeTFT with various drain bias (V_d). b) Potential profiles at ferroelectric–semiconductor interface during the forward (solid curves) and backward (dashed curves) scans of a BCBG FeTFT. The curves with the same color correspond with the same drain current which are marked beside them.

our simulated results are 9.6 V (V_{th+}) and −9.6 V (V_{th-}) for forward scan and backward scan, respectively. The step of voltage sweep was set to 0.1 V. This interval is small enough to choose the actual state according to current from transfer curves in Figure 2a, even in the subthreshold region. Note that the transfer characteristic curve in Figure 2a shows no characteristic of a p-type transistor. Under the negative bias, the holes are accumulated near the ferroelectric layers. But the electrode is heavily doped with donor, so there is a barrier that blocks holes' transport. The potential curves in Figure 2b show this scenario. Thus, the transfer curves do not show a p-type behavior at negative V_g.

3. Results and Discussion

3.1. The Electric Field in the Space Charge Region Near the Drain Edge

Using Equation (1), the electric potential can accurately describe the overall state of charge accumulation. So, the potential profiles provide clues about the condition of carriers,[25,42,43] like the energy barrier and driving force for charge drifting. Thus, the potential curves at the ferroelectric–semiconductor interface with same drain current in the forward and backward scan are presented in Figure 2b. At the beginning of the backward scan (black dash curve in Figure 2b, which is overlapped with the black solid curve), the potential distributes linearly along the channel. This is the strong inversion regime of the semiconductor layer. While sweeping V_g below V_{th}, the semiconducting layer turns into an accumulation regime and holes become the majority carriers in the channel. At the same time, the energy barrier at the drain will stop the holes from moving. As the voltage decrease, the space charge region will gradually be narrowed, and the lateral electric field is enhanced.

Figure 3a presents the lateral component of the electric field at the ferroelectric–semiconductor interface in the vicinity of the drain junction under various negative V_g. While sweeping the V_g to V_{gmax-}, the space charge region narrows, and the electric field is enhanced. The peak of the electric field in this region reaches 3.45 MV cm^{-1}. Since the tangential component of the electrostatic field is continuous across the interface, this electric field will penetrate the ferroelectric layer.

Figure 3b shows that this electric field penetrates about 20 nm into the ferroelectric layer for both near the source and drain edge while V_g at V_{gmax-}. The magnitude of the lateral electric field at the drain edge is always higher than which near the source edge in Figure 3b. The drain junction is reversely biased while the FeTFT is operated to off-state, so the voltage drop is concentrated near the drain electrode for the off-state. This can be confirmed by the magenta curve in Figure 2b.

Figure 3c confirms that while V_g approaching V_{gmax-}, the vertical component near both drain and source edges is much lower than that in the center of the channel. Figure 3d shows the vectors of the electric field in the ferroelectric layer near the edge of the drain electrode while V_g at V_{gmax-}. It shows that the electric field near the drain edge is concentrating in the direction along the channel.

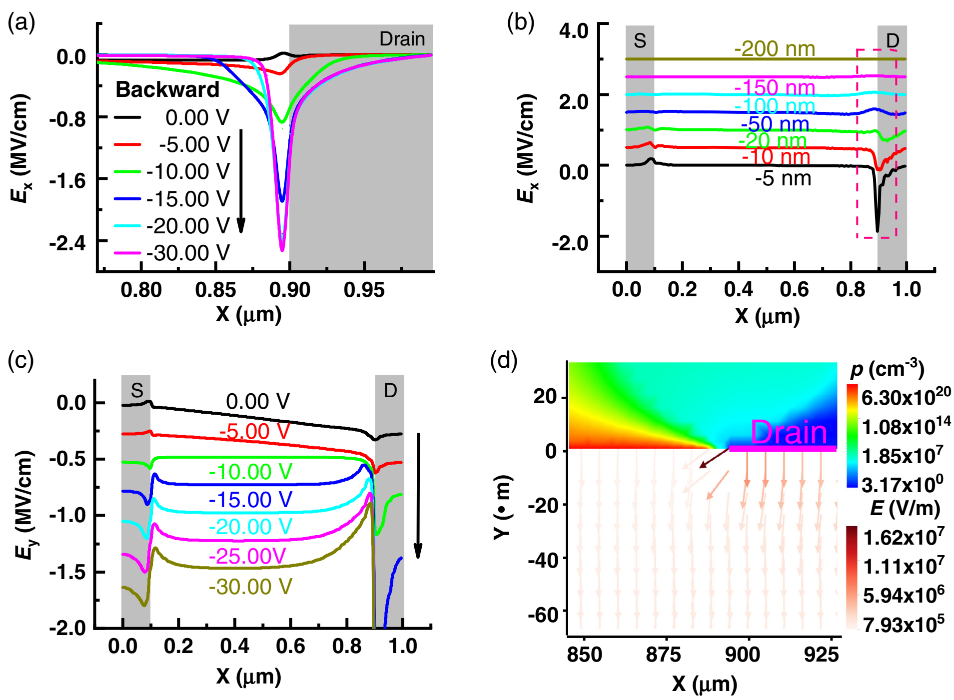

Figure 3. a,b) The lateral (E_x) and vertical (E_y) components of the electric field at the ferroelectric–semiconductor interface in the vicinity of the electrode in backward scan under various negative V_g. The curves in (b) are shifted vertically for clarity. c) The vertical component of the electric field in the various depth of the ferroelectric layer for V_g at V_{gmax-}. The gray shadows in the right and left sides of (a), (b), and (c) are the position of drain and source electrodes, respectively. d) The color map of holes density in the semiconductor layer and vector distribution of electric field in the ferroelectric layer in the vicinity of drain contact for V_g at V_{gmax-}. The black heavy line indicates the position of drain electrode. The drain contact was biased at +5 V.

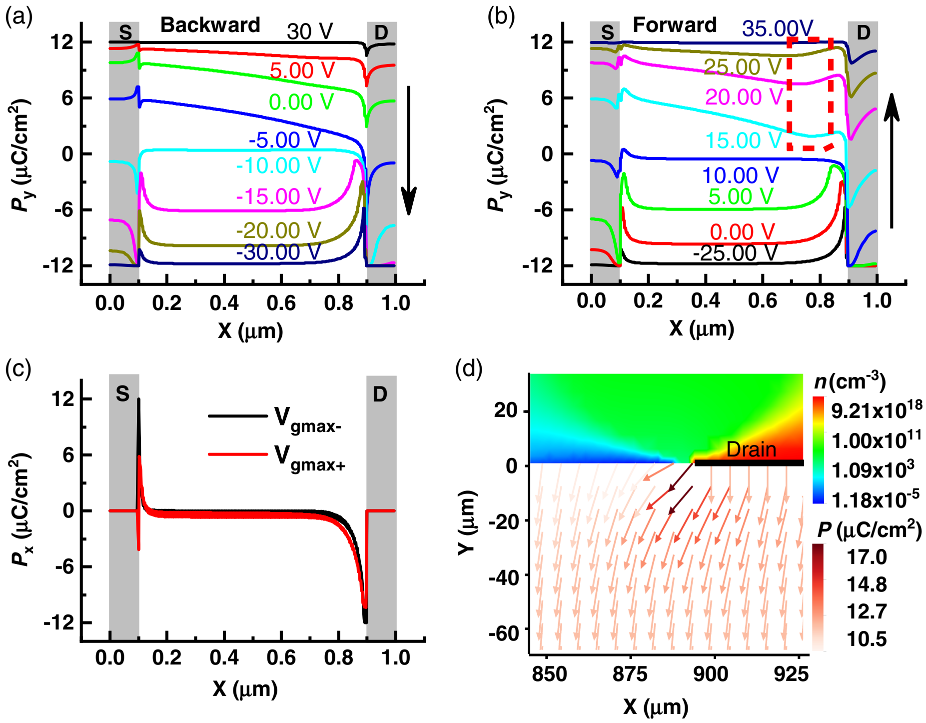

Figure 4. a,b) The vertical component of ferroelectric polarization (P_y) at ferroelectric–semiconductor interface under various V_g in backward and forward scans. c) The lateral component of ferroelectric polarization (P_x) for V_g at V_{gmax+} and V_{gmax+}. The gray shadows on both left and right sides in (a), (b), and (c) indicate the position of source (S) and drain (D) electrodes. d) The color map of electrons density in the semiconductor layer and the vector distribution of polarization in the ferroelectric layer in the vicinity of drain contact for V_g at V_{gmax-}. The black heavy line is the position of the drain electrode. The drain contact was biased at +5 V.

Since the electric field drives dipoles, its orientation and magnitude will determine the distribution of polarization in the ferroelectric layer. **Figure 4** exhibits the vertical component of polarization at the ferroelectric–semiconductor interface in both backward and forward scans.

In Figure 4a, while sweeping V_g from V_{gmax+}, the curves shows a straight-line sloping to bottom right. This is due to a positive bias applied on drain contact. While V_g increases to V_{gmax-}, the values of the curve turn to be negative. But the vertical component near the drain electrode is much weaker than that in the center of the channel. This is in correspondence with the electric field distribution presented in Figure 3. While sweeping V_g in the forward scan, the curves in Figure 4b present that the lowest point of the vertical component of polarization is about 150 nm away from the drain edge while $V_g > V_{th+}$. For the curve of V_g at 15 V in Figure 4b, The maximum value of vertical polarization near the drain edge is 2.33 μC cm^{-2}, and the minimum value in the channel is 1.90 μC cm^{-2}. Figure 4c shows that the lateral component of polarization near the edges of the source and drain is much large than that in the center of the channel (in the red dash box of Figure 4c). In Figure 4d, the vector of polarization in the vicinity of drain contact is mainly along the lateral direction to the channel side.

Laudari et al. have reported that the FeTFT presented high subthreshold swing values and low leakage gate current when the dielectric layer was laterally poled in off-state.[26] From our analysis, we know that the polarization in this region is laterally poled in off-state (Figure 4c,d), although those regions a little far from both source and drain electrodes are vertically poled. To achieve a quantitative analysis for the consequence of this phenomenon, we should focus on the simulation details in Equation (3).

From the literature, c_{off_y} and c_y in Equation (3), are two parameters to describe the polarization hysteresis loop in vertical direction.[36,44] For a fully polarized hysteresis loop, the value of c_{off_y} is 0, that is, the black lines in Figure 1c. The red line in Figure 1c presents the partly polarized hysteresis loop, in which the value of c_{off_y} is positive. In our simulation process, the value of c_y and c_{off_y} were iterated only when the sweep direction of vertical electric field changes. While sweeping the V_g in the forward scan, the value of c_{off_y} has determined at the beginning when V_g is V_{gmax-}. Thus, the value of c_{off_y} denotes the history of the polarization field. The larger the value of c_{off_y}, the more dipoles oriented along the direction of vertical electric field change. So, c_{off_y} represents the ratio of dipoles which has been aligned with the orientation of the electric field scanning direction. The color map below $y = 0$ in **Figure 5**a shows the value of c_{off_y} near the edge of drain in forward scan. It presents that a large ratio of dipoles has already oriented along the positive direction of the Y-axis at the beginning of the forward scan. Afterward, when V_g is swept to V_{gmax+}, the channel will change from the accumulation regime to the inversion regime. At the same time, the polarization of this region should be stronger than other parts in the

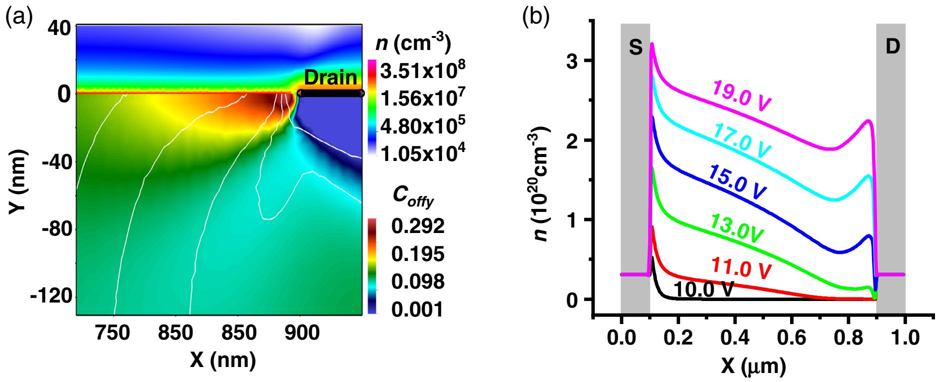

Figure 5. a) The color map of electron density and c_{off_y}. The upper right corner is the location of the drain. b) Electron density in ferroelectric–semiconductor interface for various V_g in the forward scan. The black heavy line is the position of the drain electrode. The drain contact was biased at +5 V.

channel. The curves for $V_g > V_{th+}$ in Figure 4b confirm this judgment, because the lowest position of vertical polarization at the ferroelectric–semiconductor interface does not locate at the edge of the drain electrode.

At such conditions, the device is turning to the inversion regime, and the polarization at the interface should be compensated by electrons. Thus, in Figure 5b, the electron density near the edge of the source and drain contacts are higher than other parts in the channel. As the concentration of carriers is inversely proportional to the electrical resistivity, the potential drop at the area with higher electron density will be smoother. So, in Figure 2b, the curves for the forward scan near the drain are above the blank shapes for the backward scan with the same magnitude of drain current. In the surface potential profiles reported by Kam et al.,[25] the potential curves close to the source and drain contacts changed more abruptly than other parts of the channel in the forward scan from off-state.

It should be noted in their report that the potential drop was not fixed at the edge of the electrode but drifted to the left while sweeping V_g from 0 to -6 V.[25] While they scan the potential profiles from the intermediate state, such a phenomenon was extremely weak. This is consistent with our simulated results in Figure 2b. Although their research was based on top contact bottom gate architecture, the lacking of electron injection in p-type organic material will expand the space charge region and make the polarization near the edge of source and drain contacts laterally poled in off-state. So, our analysis provides a more complete view.

3.2. Depolarization Field in the Thin-Film BCBG FeTFT

For an ideal ferroelectric capacitor, the dipole-induced electric field should be canceled by compensating charges on electrodes, resulting in a zero depolarization field. However, there is always a space-charge layer in the electrode nearby the ferroelectric layer, which leads to an imperfect screening effect and a depolarization field.[15,16] This phenomenon will be amplified while semiconductor acts as compensating material,[45] then makes the depolarization field become one of the main factors to determine the retention performance of FeTFTs.[17]

As the FeTFTs are utilized in multistate memory devices,[3–6,46–48] we now look at the depolarization field in various standby states. To achieve the standby condition, all contact of FeTFT should be short-circuited or applied with 0 V. For the simulation of the depolarization field, the sweeping of V_g was stop at different intermediate states in the forward scan or in the backward scan. Then, we operated the device to stand-by condition by sweeping the V_g and V_d to 0 V, successively.

After such operations, the device will turn to on-state for V_g stopped at the bias large than V_{th+}, and off-state for V_g stopped at the bias lower than V_{th-}. **Figure 6**a shows the depolarization field operated from the forward scan. The intensity of the depolarization field near the edge of the drain electrode reaches 0.15 MV cm^{-1} for V_g stopped at 13.0 V in the forward scan, which is stronger than that in the middle of the channel and is also stronger than that in the same region for the off-states (open shapes in Figure 6b). In Figure 6b, the maximum intensity of depolarization field operated from backward scan for on-state is just 0.013 MV cm^{-1} (black line), which is an order of magnitude smaller than from the forward scan.

In the standby condition, the degree of semiconductor inversion in the channel is determined by the polarization. Fermi–Dirac distribution makes the concentrations of electrons and holes present an exponentially varying property. Thus, in the process of depolarization of the on-state, the conductance in the vicinity of the drain electrode will be affected obviously. Since the resistance on the channel is connected in series, the depolarizing of dipoles in this region will turn the charge transfer mechanism from the space-charge limited regime[49] to the injection limited regime.[50]

From the transfer curves in Figure 2a, the characteristic of the hysteresis loop indicates that the same current can be achieved by sweeping the gate bias from the forward or backward scan. Different from the curves for the forward scan in Figure 4b, the curves of vertical polarization for the backward scan in Figure 4a show a monotonous decline near the edge of the drain $V_g > V_{th-}$. This lead to the depolarization field in this region for the FeTFT operated from the forward scan (Figure 6a) larger than that operated from the backward scan (Figure 6b). So, the on-state current will drop more significantly for the FeTFT operated

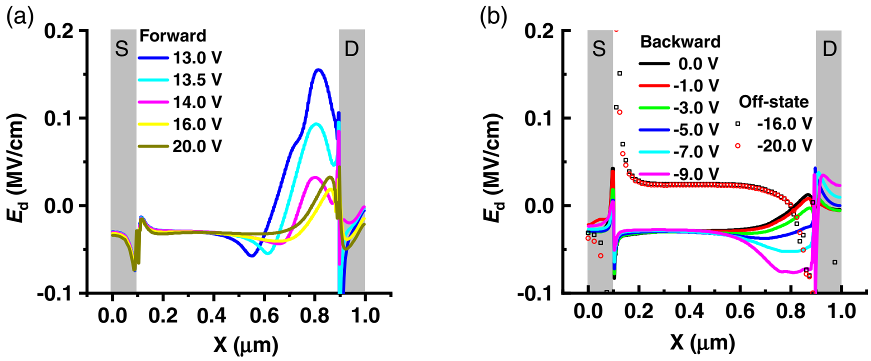

Figure 6. The depolarization field (E_d) at ferroelectric–semiconductor interface in different standby states operated from the forward scan (a) and backward scan (b). The final bias of V_g in the forward and backward scan are marked in the legend. The open shapes are the depolarization field in the off-state.

from the forward scan. This result suggests that sweeping the gate bias in the backward scan from the on-state is a better choice if the BCBG FeTFT is employed as a multi-state memory device.

For the off-state, although the right side of the depolarization field in the off-state also presents a spine near the source edge (Figure 6b), most of the channel is still in the accumulation regime. Low electron concentration in the channel will keep the device stay in the off-state. Hence, the off-state exhibits a better retention performance than the on-state.

The leakage current of the ferroelectric layer is also an important factor affecting the retention characteristics of FeTFT.[18,51] The leakage current is driven by the depolarization field of the ferroelectric layer in stand-by conditions. So, the leakage current in FeTFT can be effectively reduced by turning the polarization direction from vertical to lateral.

Laudari et al.[26] has demonstrated that the lateral poled ferroelectric insulator in organic FeTFT presented reduced the gate leakage current. From our simulated results, the off-state of BCBG FeTFT inherently possesses such characteristics near the edge of the drain electrode. Hence, from the perspective of leakage, the off-state will also show better retention characteristics than the on-state. Huang et al.[17] reported the opposite result from a different perspective. They believed that the accumulation regime of off-state enhanced remnant polarization and neglected the contribution of V_d. However, our analysis is based on the electrode details and leakage current in the FeTFT.

4. Conclusion

In this article, we demonstrated the use of TCAD simulation for n-type FeTFT devices. The distribution of physical fields in the BCBG FeTFT was simulated, and the results reveal that the junction field near the drain electrode plays a critical role in the electrical performance of FeTFT. The polarization in the vicinity of the drain electrode is pinned along channel length direction by the junction field in off-state. The leakage gate current in off-state should be reduced. The vertical depolarization filed in the same region while operating the device in the forward scan to on-state is enhanced. The off-state of BCBG FeTFT should present better retention performance compared with the on-state. If this kind of FeTFT is deployed as a multi-state memory device, different states should be realized in the backward scan from on-state rather than in the forward scan from off-state. Based on the Boltzmann approximation of semiconductors, the electrons and holes will show symmetrical physical behaviors. So, the depolarization field of the p-type FeTFTs will present the same characteristics as the n-type FeTFTs.

Acknowledgements

This work was financially supported by the Science and Technology Commission of Shanghai Municipality (STCSM, Grant Nos. 21520711600 and 20ZR1408200), the National Natural Science Foundation of China (NSFC, Grant No. 12174061), Guizhou Provincial Science and Technology Projects (Grant Nos. QKHJC-ZK [2021] YB329 and QKHJC-ZK [2021] YB018), and the Fujian Province Young and Middle-aged Teacher Education Research Project (Grant No. JAT200256)

Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

Research data are not shared.

Keywords

depletion layer electric field, depolarization field, ferroelectric thin-film transistors, leakage current, simulation

Received: June 27, 2021
Revised: September 1, 2021
Published online: October 13, 2021

[1] T. Ali, P. Polakowski, S. Riedel, T. Büttner, T. Kämpfe, M. Rudolph, B. Pätzold, K. Seidel, D. Löhr, R. Hoffmann, M. Czernohorsky, K. Kühnel, P. Steinke, J. Calvo, K. Zimmermann, J. Müller, *IEEE Trans. Electron Devices* **2018**, *65*, 3769.
[2] W. Xiao, C. Liu, Y. Peng, S. Zheng, Q. Feng, C. Zhang, J. Zhang, Y. Hao, M. Liao, Y. Zhou, *Nanoscale Res. Lett.* **2019**, *14*, 1.

[3] A. K. Tripathi, A. J. J. M. van Breemen, J. Shen, Q. Gao, M. G. Ivan, K. Reimann, E. R. Meinders, G. H. Gelinck, *Adv. Mater.* **2011**, *23*, 4146.
[4] Y. You, K. Yang, S. Yuan, S. Dong, H. Zhang, Q. Huang, W. P. Gillin, Y. Zhan, L. Zheng, *Org. Electron.* **2014**, *15*, 1983.
[5] D. Lee, S. M. Yang, T. H. Kim, B. C. Jeon, Y. S. Kim, J.-G. Yoon, H. N. Lee, S. H. Baek, C. B. Eom, T. W. Noh, *Adv. Mater.* **2012**, *24*, 402.
[6] S. K. Hwang, I. Bae, R. H. Kim, C. Park, *Adv. Mater.* **2012**, *24*, 5910.
[7] R. C. G. Naber, C. Tanase, P. W. M. Blom, G. H. Gelinck, A. W. Marsman, F. J. Touwslager, S. Setayesh, D. M. de Leeuw, *Nat. Mater* **2005**, *4*, 243.
[8] H.-D. Wang, D. K. Sang, Z.-N. Guo, R. Cao, J.-L. Zhao, M. N. U. Shah, T.-J. Fan, D.-Y. Fan, H. Zhang, *Chin. Phys. B* **2018**, *27*, 087308.
[9] Y. Xu, J. Yuan, K. Zhang, Y. Hou, Q. Sun, Y. Yao, S. Li, Q. Bao, H. Zhang, Y. Zhang, *Adv. Funct. Mater.* **2017**, *27*, 1702211.
[10] S. C. Dhanabalan, J. S. Ponraj, Z. Guo, S. Li, Q. Bao, H. Zhang, *Adv. Sci.* **2017**, *4*, 1600305.
[11] D. K. Sang, H. Wang, Z. Guo, N. Xie, H. Zhang, *Adv. Funct. Mater.* **2019**, *29*, 1903419.
[12] J. Íñiguez, P. Zubko, I. Luk'yanchuk, A. Cano, *Nat. Rev. Mater.* **2019**, *4*, 243.
[13] M. Si, C. Jiang, W. Chung, Y. Du, M. A. Alam, P. D. Ye, *Nano Lett.* **2018**, *18*, 3682.
[14] R. Khosla, S. K. Sharma, *ACS Appl. Electron. Mater.* **2021**, *3*, 2862.
[15] R. R. Mehta, B. D. Silverman, J. T. Jacobs, *J. Appl. Phys.* **1973**, *44*, 3379.
[16] P. Wurfel, I. P. Batra, *Ferroelectrics* **1976**, *12*, 55.
[17] S. Huang, X. Zhong, Y. Zhang, Q. Tan, J. Wang, Y. Zhou, *IEEE Trans. Electron Devices* **2011**, *58*, 3388.
[18] X. Pan, T. P. Ma, *Appl. Phys. Lett.* **2011**, *99*, 013505.
[19] Z. Hu, Q. Feng, Z. Feng, Y. Cai, Y. Shen, G. Yan, X. Lu, C. Zhang, H. Zhou, J. Zhang, Y. Hao, *Nanoscale Res. Lett.* **2019**, *14*, 1.
[20] S. Kim, V. Gopalan, A. Gruverman, *Appl. Phys. Lett.* **2002**, *80*, 2740.
[21] S. Ducharme, V. M. Fridkin, A. V. Bune, S. P. Palto, L. M. Blinov, N. N. Petukhova, S. G. Yudin, *Phys. Rev. Lett.* **2000**, *84*, 175.
[22] K. Asadi, P. W. M. Blom, D. M. de Leeuw, *Appl. Phys. Lett.* **2011**, *99*, 053306.
[23] Z. Fu, J. Zhang, J. Weng, W. Chen, Y. Jiang, G. Zhu, *AIP Adv.* **2015**, *5*, 097211.
[24] R. Cai, A. M. Jonas, *Sci. Rep.* **2016**, *6*, 1.
[25] B. Kam, X. Li, C. Cristoferi, E. C. P. Smits, A. Mityashin, S. Schols, J. Genoe, G. Gelinck, P. Heremans, *Appl. Phys. Lett.* **2012**, *101*, 033304.
[26] A. Laudari, A. Pickett, F. Shahedipour-Sandvik, K. Hogan, J. E. Anthony, X. He, S. Guha, *Adv. Mater. Interfaces* **2019**, *6*, 1801787.
[27] A. D. Hilton, C. A. Randall, D. J. Barber, T. R. Shrout, *Ferroelectrics* **1989**, *93*, 379.
[28] S. Selberherr, *Analysis and Simulation of Semiconductor Devices*, Springer-Verlag, New York **1984**.
[29] A. Laudari, S. Guha, *Phys. Rev. Appl.* **2016**, *6*, 4.
[30] M. Si, A. K. Saha, S. Gao, G. Qiu, J. Qin, Y. Duan, J. Jian, C. Niu, H. Wang, W. Wu, S. K. Gupta, P. D. Ye, *Nat Electron* **2019**, *2*, 580.
[31] J. Yang, P.-Y. Liao, T.-C. Chang, B.-W. Chen, H.-C. Huang, H.-C. Chiang, W.-C. Su, Q. Zhang, *IEEE Electron Device Lett.* **2017**, *38*, 592.
[32] L. Tu, R. Cao, X. Wang, Y. Chen, S. Wu, F. Wang, Z. Wang, H. Shen, T. Lin, P. Zhou, X. Meng, W. Hu, Q. Liu, J. Wang, M. Liu, J. Chu, *Nat. Commun.* **2020**, *11*, 101.
[33] J. Yang, W. Li, D. Wang, Y. Li, *Small Struct.* **2021**, *2*, 2000051.
[34] X. Wang, P. Wang, J. Wang, W. Hu, X. Zhou, N. Guo, H. Huang, S. Sun, H. Shen, T. Lin, M. Tang, L. Liao, A. Jiang, J. Sun, X. Meng, X. Chen, W. Lu, J. Chu, *Adv. Mater.* **2015**, *27*, 6575.
[35] W. J. Hu, D.-M. Juo, L. You, J. Wang, Y.-C. Chen, Y.-H. Chu, T. Wu, *Sci. Rep.* **2014**, *4*, 1.
[36] M. Ghittorelli, T. Lenz, H. S. Dehsari, D. Zhao, K. Asadi, P. W. M. Blom, Z. M. Kovács-Vajna, D. M. de Leeuw, F. Torricelli, *Nat Commun* **2017**, *8*, 1.
[37] *DEVSIM LLC, DEVSIM TCAD semiconductor device simulator*, https://devsim.org (accessed: June 2021).
[38] C. Geuzaine, J.-F. Remacle, *Int. J. Numer. Meth. Eng.* **2009**, *79*, 1309.
[39] S. L. Miller, R. D. Nasby, J. R. Schwank, M. S. Rodgers, P. V. Dressendorfer, *J. Appl. Phys.* **1990**, *68*, 6463.
[40] B. Shrestha, R. Pieper, W. Wondmagegn, N. Satyala, in *2011 Proc. IEEE Southeastcon*, IEEE, Piscataway, NJ **2011**.
[41] Z.-Y. Fu, J.-C. Zhang, J.-H. Hu, Y.-L. Jiang, S.-J. Ding, G.-D. Zhu, *Chin. Phys. B* **2015**, *24*, 058502.
[42] V. Kaphle, P. R. Paudel, D. Dahal, R. K. R. Krishnan, B. Lüssem, *Nat. Commun.* **2020**, *11*, 1.
[43] E. C. P. Smits, S. G. J. Mathijssen, M. Cölle, A. J. G. Mank, P. A. Bobbert, P. W. M. Blom, B. de Boer, D. M. de Leeuw, *Phys. Rev. B* **2007**, *76*, 12.
[44] B. Jiang, P. Zurcher, R. Jones, S. Gillespie, J. Lee, in *1997 Symp. on VLSI Technology*, IEEE, Piscataway, NJ **1997**, pp. 141–142.
[45] S. Majumdar, B. Chen, Q. H. Qin, H. S. Majumdar, S. van Dijken, *Adv. Funct. Mater.* **2017**, *28*, 1703273.
[46] H. Mulaosmanovic, T. Mikolajick, S. Slesazeck, *Appl. Mater. Interfaces* **2018**, *10*, 23997.
[47] D. Zhao, I. Katsouras, K. Asadi, W. A. Groen, P. W. M. Blom, D. M. de Leeuw, *Appl. Phys. Lett.* **2016**, *108*, 232907.
[48] Y. Kaneko, Y. Nishitani, M. Ueda, E. Tokumitsu, E. Fujii, *Appl. Phys. Lett.* **2011**, *99*, 182902.
[49] V. D. Mihailetchi, J. Wildeman, P. W. M. Blom, *Phys. Rev. Lett.* **2005**, *94*, 12.
[50] U. Wolf, V. I. Arkhipov, H. Bässler, *Phys. Rev. B* **1999**, *59*, 7507.
[51] T. Ma, J.-P. Han, *IEEE Electron Device Lett.* **2002**, *23*, 386.

